WASP-11b

R-1430

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_191027` · created 2026-08-01T07:22:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458779.9693 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.051
Transit depth (Rp/Rs)^2	3.05 +/- 1.79 [%]
Orbital Inclination (inc)	85.73 +/- 1.73
Airmass coefficient 1 (a1)	1.6 +/- 1.03
Airmass coefficient 2 (a2)	-0.18 +/- 0.13
Scatter in the residuals of the lightcurve fit is	55.12 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	3.75
Transit Duration (day)	0.012 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.051 vs 0.131 (deviation 0.86σ)
Duration fit vs published	0.012 d vs 0.1075 d (deviation 8.68σ)
Mid-time O–C	-5354.0 min
Depth SNR	3.4
Residual scatter	55.12 %

Light curve

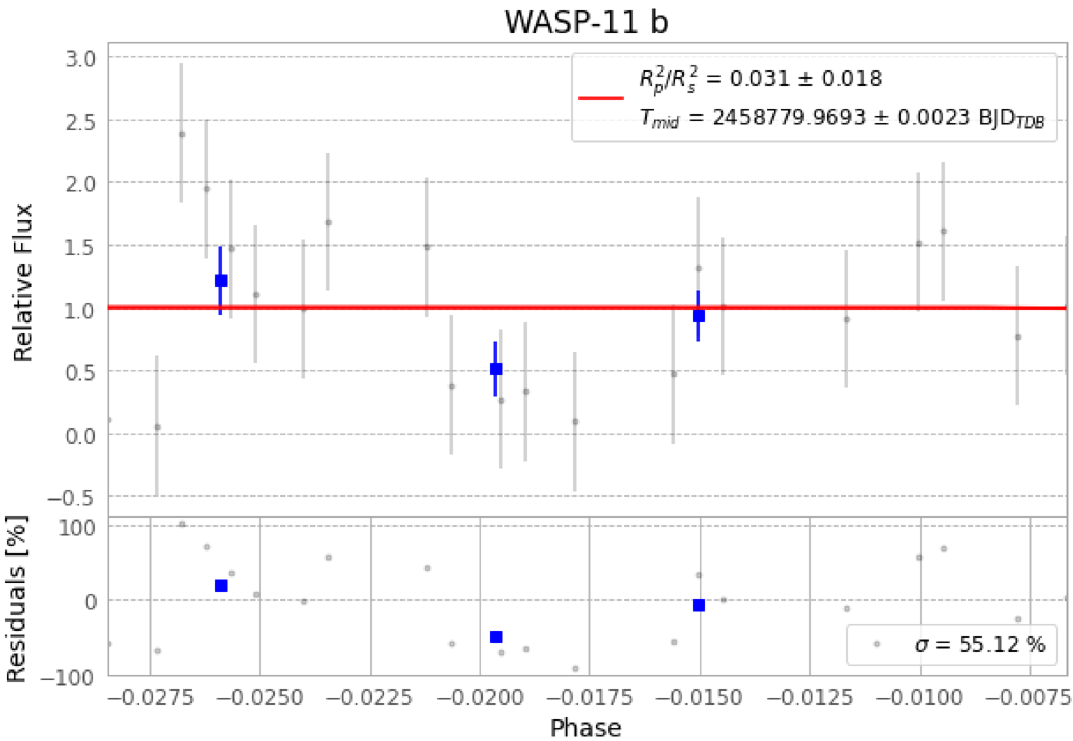

Figure 1 — FinalLightCurve_WASP-11 b_2019-10-26.png (SHA-256 582f3657bea3d2f8…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [551, 136], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 da510deb4a8d48d2…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

94 FITS frames (62 MB), `HATP-10191027015412.FITS` → `HATP-10191027063312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-191027.log (722a29bae879…), FinalLightCurve_WASP-11 b_2019-10-26.png (582f3657bea3…), FinalLightCurve_WASP-11 b_2019-10-26.csv (a65a529da9af…)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-01 07:22Z.